

Continuity of Convex Functions on $\mathbb{R}^n$

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1 Introduction

This note proves that a convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous everywhere. The result is classical. The usual argument invokes the fact that a cube is the convex hull of its 2^n vertices, and then applies Jensen's inequality in several variables.

I wrote this while in *classes préparatoires*, where convex functions of several variables and the multivariable form of Jensen's inequality are outside the syllabus. The argument below therefore uses nothing beyond the two-point definition of convexity. It builds a chain of n inequalities, each one pinning a further coordinate to ± 1 , until the point reaches a vertex of the hypercube. The idea arose while exploring how convex functions of one variable generalise to several dimensions.

One remark on which hypothesis does the work. The bounding argument of Section 3.2 only ever uses $f(\lambda a + (1 - \lambda)b) \leq \max(f(a), f(b))$, that is quasiconvexity. Full convexity is needed only in Section 3.3, where the slopes of a one-dimensional restriction are compared.

2 Preliminaries

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. Recall that f is convex if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

We aim to prove that f is continuous everywhere in $\mathbb{R}^n$.

To do this, we use the following construction.

- We define the unit sphere S in the Euclidean norm $\|\cdot\|_2$ as

$$S = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}.$$

- We construct a hypercube containing S by introducing the supremum norm

$$\|x\|_\infty = \max(|x_1|, |x_2|, \dots, |x_n|).$$

- The unit sphere in this norm is the boundary of an n -dimensional hypercube centered at 0, with vertices where each coordinate is ± 1 . In this context, we are only interested in the boundary of the hypercube. We therefore define H as the set of points satisfying $\|x\|_\infty = 1$, which ensures that at least one coordinate of x reaches ± 1 .

3 Proof of Continuity

3.1 Reduction by Contradiction

Assume, for the sake of contradiction, that f is not continuous at some point $x_0 \in \mathbb{R}^n$. We may consider the function $x \mapsto f(x + x_0)$, so we can assume $x_0 = 0$. By definition, there exists $\varepsilon > 0$ such that for any $\delta > 0$ we can find $x \in B(0, \delta)$ satisfying

$$|f(x) - f(0)| \geq \varepsilon.$$

Define the Euclidean unit sphere $S = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$. Since S is compact and f is convex, we aim to show that $f(S)$ is bounded. This will contradict the assumption of discontinuity, as shown in the last part of the proof.

3.2 Recursive Bounding of $f(S)$

We now present the recursive argument bounding $f(S)$. Define the boundary of the hypercube H as

$$H = \{x \in \mathbb{R}^n : \|x\|_\infty = 1\},$$

and let $V(H)$ denote its vertices, the points where all n coordinates are equal to ± 1 . Let $z \in S$ be a point on the unit sphere. $\|z\|_\infty \leq \|z\|_2 = 1$. The argument constructs a sequence of points y_k such that $f(z)$ is bounded by the maximum of f evaluated at the vertices of the hypercube, or at $f(y_k)$. The recursion stops when y_k becomes a vertex of the hypercube.

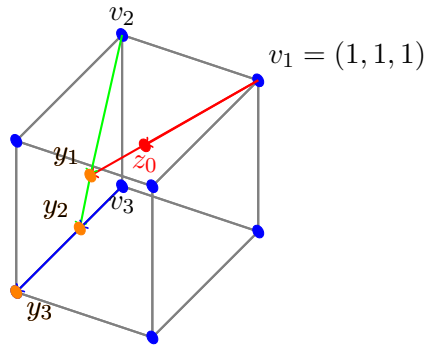


Figure 1: Illustration of the hypercube H for $n = 3$ and the progression of points y_k towards z_0 . y_1 lies on a face, y_2 on an edge, and y_3 is a vertex.

Step 1: setting. We aim to construct a sequence of points y_k such that

- at least k coordinates of y_k are equal to ± 1 ,
- $\|y_k\|_\infty = 1$,
- $f(z) \leq \max(\max_{v \in V(H)} f(v), f(y_k))$.

Step 2: base case ($k = 1$). Start by constructing the line passing through the vertex $v_0 = (1, 1, \dots, 1)$ and z . This line is parameterized as

$$\gamma(t) = v_0 + t(z - v_0), \quad t \in \mathbb{R}.$$

In terms of coordinates, the i -th coordinate of $\gamma(t)$ is

$$\gamma(t)_i = 1 + t(z_i - 1), \quad i = 1, \dots, n.$$

For each coordinate i , use the intermediate value theorem to find a (unique) $t'_i \geq 1$ such that $\gamma(t'_i)_i = \pm 1$. Indeed, $-1 \leq \gamma(1)_i = z_i \leq 1$ and $\gamma(t)_i$ diverges to $\pm\infty$ as t grows to ∞ and $t \mapsto \gamma(t)_i$ is monotone and continuous.

Define

$$t_1 = \min_i t'_i.$$

The choice $t_1 = \min_i t'_i$ ensures that the point $y_1 = \gamma(t_1)$ remains on the boundary of the hypercube H : the smallest t'_i guarantees that at least one coordinate of y_1 reaches ± 1 without the other coordinates exceeding the bounds imposed by H . By construction,

- y_1 has at least one coordinate equal to ± 1 ,
- $\|y_1\|_\infty = 1$.

Note that, because $t_1 > 1$, we have $z \in [v_0, y_1]$. By convexity of f ,

$$f(z) \leq \max(f(v_0), f(y_1)).$$

This establishes the base case.

Step 3: inductive hypothesis. Assume that for some $k \geq 1$ there exists a point $y_k \in H$ such that

- at least k coordinates of y_k are equal to ± 1 ,
- $\|y_k\|_\infty = 1$,
- $f(z) \leq \max(\max_{v \in V(H)} f(v), f(y_k))$.

Step 4: inductive step ($k \rightarrow k+1$). If y_k already has n coordinates equal to ± 1 , then y_k is a vertex of H and the proof is complete. Otherwise, there exists at least one coordinate j such that

$$y_k[j] \notin \{-1, 1\}.$$

Define y'_k by fixing to 1 all coordinates of y_k that are not ± 1 , and keeping the other coordinates unchanged. Thus y'_k is a vertex of H . Construct the line passing through y'_k and y_k , parameterized as

$$\gamma(t) = y'_k + t(y_k - y'_k), \quad t \in \mathbb{R}.$$

In terms of coordinates, the i -th coordinate of $\gamma(t)$ is

$$\gamma(t)_i = \begin{cases} y_k[i], & \text{if } y_k[i] = \pm 1, \\ 1 + t(y_k[i] - 1), & \text{otherwise.} \end{cases}$$

Using the intermediate value theorem, for each coordinate i not fixed at ± 1 , find $t'_i > 1$ such that $\gamma(t'_i)_i = \pm 1$. Define

$$t_{k+1} = \min_i t'_i.$$

The point $y_{k+1} = \gamma(t_{k+1})$ lies on the boundary of H and has at least $k+1$ coordinates equal to ± 1 . Note that, because $t_{k+1} > 1$, we have $y_k \in [y'_k, y_{k+1}]$. Thus, by convexity of f ,

$$f(y_k) \leq \max(f(y'_k), f(y_{k+1})),$$

and therefore

$$f(z) \leq \max\left(\max_{v \in V(H)} f(v), f(y_{k+1})\right).$$

Step 5: termination. The recursion terminates when $k = n$, at which point y_k is a vertex of H . Thus $f(z)$ is bounded by

$$f(z) \leq \max_{v \in V(H)} f(v).$$

Since H has a finite number of vertices, and z was chosen arbitrarily in S , the set $f(S)$ is bounded.

3.3 Contradiction and Conclusion

We have shown that $f(S)$, the image of the unit sphere S , is bounded. Without loss of generality, we may assume $f(0) = 0$, since subtracting a constant from f affects neither its convexity nor the argument.

Let $M > 0$ be a real number. Define

$$a = \min\left(0.5, \frac{\varepsilon}{M}\right).$$

By the assumption of discontinuity, there exists $y \in B(0, a)$ such that $|f(y) - f(0)| \geq \varepsilon$, which implies $|f(y)| \geq \varepsilon$.

Now, define the convex function

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(t) = f\left(\frac{t y}{\|y\|_2}\right).$$

Case 1: $f(y) \geq \varepsilon$. Using the convexity of g , consider its slopes at the points 0, $\|y\|_2$ and 1. The convexity condition gives

$$\frac{g(\|y\|_2) - g(0)}{\|y\|_2 - 0} \leq \frac{g(1) - g(0)}{1 - 0}.$$

Substituting $g(0) = f(0) = 0$, this simplifies to

$$\frac{g(\|y\|_2)}{\|y\|_2} \leq g(1).$$

Since $g(\|y\|_2) = f(y) \geq \varepsilon$ and $\|y\|_2 \leq a$, we have

$$\frac{\varepsilon}{a} \leq g(1) = f\left(\frac{y}{\|y\|_2}\right).$$

Since $a \leq \varepsilon/M$,

$$M \leq f\left(\frac{y}{\|y\|_2}\right).$$

Case 2: $f(y) \leq -\varepsilon$. Similarly, consider the slopes of g at the points -1 , 0 and $\|y\|_2$. Convexity gives

$$\frac{g(\|y\|_2) - g(0)}{\|y\|_2 - 0} \geq \frac{g(0) - g(-1)}{0 - (-1)}.$$

Substituting $g(0) = 0$, this simplifies to

$$\frac{g(\|y\|_2)}{\|y\|_2} \geq -g(-1).$$

Since $g(\|y\|_2) = f(y) \leq -\varepsilon$, we find

$$\frac{-\varepsilon}{\|y\|_2} \geq -g(-1) = -f\left(-\frac{y}{\|y\|_2}\right),$$

thus

$$\frac{\varepsilon}{\|y\|_2} \leq f\left(-\frac{y}{\|y\|_2}\right).$$

Since $\|y\|_2 \leq a \leq \varepsilon/M$,

$$M \leq f\left(-\frac{y}{\|y\|_2}\right).$$

Conclusion. In both cases we find that either

$$f\left(\frac{y}{\|y\|_2}\right) \geq M \quad \text{or} \quad f\left(-\frac{y}{\|y\|_2}\right) \geq M.$$

Since $\pm y/\|y\|_2 \in S$ and M was arbitrary, this shows that $f(S)$ is unbounded. This contradicts our earlier conclusion that $f(S)$ is bounded. Hence f is continuous at x_0 . Since x_0 was chosen arbitrarily, f is continuous everywhere in $\mathbb{R}^n$. $\square$

4 Conclusion

This proof establishes the continuity of convex functions on $\mathbb{R}^n$ using a geometric argument based on hypercubes and recursive bounding. By leveraging the hierarchical structure of the hypercube and the convexity of f , we showed that the function values on the unit sphere are bounded, and derived a contradiction from the assumption of discontinuity.